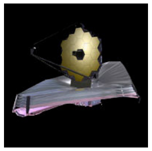

5114 - Tracing the evolution of circumstellar and protoplanetary disks at low metallicity

Cycle: 3, Proposal Category: GO

INVESTIGATORS

Name	*Institution*
Dr. Elena Sabbi (PI)	**NOIRLab - Gemini North (HI)**
Dr. Peter Zeidler (CoI) (ESA Member) (CoPI)	Space Telescope Science Institute - ESA - JWST
Dr. Beena Meena (CoI)	Space Telescope Science Institute
Dr. Guido De Marchi (CoI) (ESA Member)	European Space Agency - ESTEC
Dir. Margaret Meixner (CoI)	Jet Propulsion Laboratory
Dr. Antonella Nota (CoI)	Space Telescope Science Institute
Dr. Remy Indebetouw (CoI)	The University of Virginia
Mx. Theo O'Neill (CoI)	Center for Astrophysics | Harvard & Smithsonian
Petia Yanchulova Merica-Jones (CoI)	Space Telescope Science Institute
Dr. Olivia Jones (CoI) (ESA Member)	United Kingdom Astronomy Technology Centre
Dr. James Muzerolle Page (CoI)	Space Telescope Science Institute
Dr. Isha Nayak (CoI)	NASA Goddard Space Flight Center
Nolan Habel (CoI)	Jet Propulsion Laboratory
Dr. Laurie E Chu (CoI)	Caltech/IPAC
Dr. Alec S. Hirschauer (CoI)	Space Telescope Science Institute
Burcu Gunay (CoI)	Space Telescope Science Institute
Dr. Suzanne Madden (CoI) (ESA Member)	CEA/DSM/DAPNIA/Service d'Astrophysique
Dr. Karl D. Gordon (CoI)	Space Telescope Science Institute
Dr. Laura Lenkic (CoI)	California Institute of Technology
Dr. Tony Wong (CoI)	University of Illinois at Urbana - Champaign
Dr. Massimo Robberto (CoI)	Space Telescope Science Institute

Name	*Institution*
Mr. Jeroen Jaspers (CoI) (ESA Member)	Dublin Institute For Advanced Studies

OBSERVATIONS

Folder	*Observation*	*Label*	*Observing Template*	*Science Target*
Observation Folder				
	3	N159-OFF NIRCam	NIRCam Imaging	(7) OFF-FIELD-NIRCam
	10	N159 NIRCam	NIRCam Imaging	(5) LHA-120-N-159-NIRCam
	11	N159S NIRCam	NIRCam Imaging	(9) LHA-120-N-159S-NIRCam
	12	N159-MIRI	MIRI Imaging	(8) LHA-120-N-159-MIRI
	22	N159-MIRI	MIRI Imaging	(10) LHA-120-N-159-MIRI-12.2-repeat
	9	N159S-MIRI	MIRI Imaging	(6) LHA-120-N-159S-MIRI
	7	N159-OFF MIRI	MIRI Imaging	(4) OFF-FIELD-MIRI
	17	N159-OFF MIRI	MIRI Imaging	(4) OFF-FIELD-MIRI

ABSTRACT

The interaction between pre-supernova FUV feedback from massive OB stars and radiation from the host stars plays a crucial, albeit not yet fully characterized, role in the evolution of the interstellar medium and circumstellar disks, as well as in the ability of low-mass stars to retain planetary systems. The synergy between NIRCam and MIRI observations finally offers us the opportunity to advance the study of the rapid (< 3 Myr) evolution of young stellar objects and pre-main sequence stars’ disks in extragalactic star-forming regions.

Our neighboring low-metallicity galaxy, the Large Magellanic Cloud, hosts the well-studied giant HII complex N159, which comprises three giant molecular clouds at various evolutionary stages. This system serves as a particularly rare example of the transition from an embedded region at the onset of high-mass star formation to a region where pre-supernova feedback has already begun to clear the natal molecular cloud, revealing massive star clusters. N159 provides a unique window for witnessing the full evolution of a giant molecular cloud under the influence of emerging star formation in conditions significantly different from those observed in the Milky Way. We have, therefore. the opportunity to document how the progressively increasing stellar feedback affects the evolution of protoplanetary disks and the properties of the ISM at low metallicity, as the FUV radiation from O and B stars clears the surrounding natal material.

OBSERVING DESCRIPTION

We request observations of the N159 complex in the Large Magellanic Clouds with NIRCam and MIRI.

For NIRCam we will use the broad, medium and narrowband filters F115W, F140M, F187N, F210M, F212N, F300M, F335M, F360M, F460M, and F470N to characterize the photosphere of young stellar objects and low-mass pre-main sequence stars.

We will use a 2x3 mosaic with 70% overlap in columns to fill the detector gap and maximize the exposure time.

For MIRI we will use a 1x4 mosaic to cover N159 East and West. A second 1x2 mosaic will be used to cover N159S. We will refine the column shift once the observing .window is defined, to maximize the covered area.

In addition we will collect observations with NIRCam and MIRI off the molecular ridge, for statistic decontamination.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(4)	OFF-FIELD-MIRI	RA: 05 41 52.0000 (85.4666667d) Dec: -69 38 12.00 (-69.63667d) Equinox: J2000		
Comments: Category=Stellar Cluster Description=[OB associations, Protoclusters, T associations, Young star clusters]				
(5)	LHA-120-N-159-NIRCam	RA: 05 39 52.3800 (84.9682500d) Dec: -69 45 42.50 (-69.76181d) Equinox: J2000	Epoch of Position: 2000	
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Stellar Cluster Description=[OB associations, Protoclusters, Young star clusters]				
(6)	LHA-120-N-159S-MIRI	RA: 05 40 4.7000 (85.0195833d) Dec: -69 50 30.60 (-69.84183d) Equinox: J2000	Epoch of Position: 2000	
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Stellar Cluster Description=[Protoclusters, Young star clusters]				
(7)	OFF-FIELD-NIRCam	RA: 05 41 52.0000 (85.4666667d) Dec: -69 38 12.00 (-69.63667d) Equinox: J2000		
Comments: Category=Stellar Cluster Description=[OB associations, Protoclusters, T associations, Young star clusters]				
(8)	LHA-120-N-159-MIRI	RA: 05 39 44.9700 (84.9373750d) Dec: -69 45 58.60 (-69.76628d) Equinox: J2000	Epoch of Position: 2000	
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Stellar Cluster Description=[OB associations, Protoclusters, Young star clusters]				
(9)	LHA-120-N-159S-NIRCam	RA: 05 40 4.7000 (85.0195833d) Dec: -69 50 30.60 (-69.84183d) Equinox: J2000	Epoch of Position: 2000	
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Stellar Cluster Description=[Protoclusters, Young star clusters]				
(10)	LHA-120-N-159-MIRI-12.2-repeat	RA: 05 39 48.7000 (84.9529167d) Dec: -69 46 15.80 (-69.77106d) Equinox: J2000	Epoch of Position: 2000	
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Stellar Cluster Description=[OB associations, Protoclusters, Young star clusters]				

Observation

Proposal 5114, Observation 3: N159-OFF NIRCam

Wed Sep 11 22:00:20 GMT 2024

Diagnostic Status: Warning

Observing Template: NIRCam Imaging

Diagnostics

(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(7)	OFF-FIELD-NIRCam	RA: 05 41 52.0000 (85.4666667d) Dec: -69 38 12.00 (-69.63667d) Equinox: J2000		

Comments:
Category=Stellar Cluster
Description=[OB associations, Protoclusters, T associations, Young star clusters]

Template

Module	Subarray	Target Placement
ALL	FULL	Module Gap

Dithers

#	Primary Dither Type	Primary Dithers	Subpixel Dither Type	Dither Size	Subpixel Positions
1	NONE		SMALL-GRID-DITHER		4

Spectral Elements

#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	ETC Wkbk.Calc ID
1	F187N	F460M	BRIGHT2	5	1	4	4	429.471	
2	F210M	F335M	BRIGHT2	5	1	4	4	429.471	
3	F140M	F360M	BRIGHT2	5	1	4	4	429.471	
4	F115W	F300M	BRIGHT2	5	1	4	4	429.471	

Special Requirements

Offset -75.0 arcsec, 0.0 arcsec

Observation

Proposal 5114, Observation 10: N159 NIRCam

Wed Sep 11 22:00:20 GMT 2024

Diagnostic Status: Warning

Observing Template: NIRCam Imaging

Diagnostics

(Visit 10:1) Warning (Form): Data Excess over lower threshold
(Visit 10:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.
(Visit 10:2) Warning (Form): Data Excess over lower threshold
(Visit 10:2) Warning (Form): Overheads are provisional until the Visit Planner has been run.
(Visit 10:3) Warning (Form): Data Excess over lower threshold
(Visit 10:3) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(5)	LHA-120-N-159-NIRCam	RA: 05 39 52.3800 (84.9682500d) Dec: -69 45 42.50 (-69.76181d) Equinox: J2000	Epoch of Position: 2000	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.
Category=Stellar Cluster
Description=[OB associations, Protoclusters, Young star clusters]

Template

Module	Subarray	Target Placement
ALL	FULL	Module Gap

Mosaic

Rows	Columns	Row Overlap %	Column Overlap %	Row shift (deg)	Column shift (deg)	Tile Order
3	1	10.0	84.0	5.0	5.0	DEFAULT

Dithers

#	Primary Dither Type	Primary Dithers	Subpixel Dither Type	Dither Size	Subpixel Positions
1	FULLBOX	4TIGHT	SMALL-GRID-DITHER		2

Spectral Elements

#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	ETC Wkbk.Calc ID
1	F212N	F470N+F444W	BRIGHT2	6	1	8	8	1030.73	
2	F187N	F460M	BRIGHT2	5	1	8	8	858.942	
3	F210M	F335M	BRIGHT2	5	1	8	8	858.942	
4	F140M	F360M	BRIGHT2	5	1	8	8	858.942	
5	F115W	F300M	BRIGHT2	5	1	8	8	858.942	

Special Requirements	
	Sequence Visits within 53.0 Days Aperture PA Range 61.92542306 to 64.92542306 Degrees (V3 62.0 to 65.0) Aperture PA Range 87.92542306 to 90.92542306 Degrees (V3 88.0 to 91.0) Aperture PA Range 130.92542306 to 198.92542306 Degrees (V3 131.0 to 199.0) Aperture PA Range 199.92542306 to 207.92542306 Degrees (V3 200.0 to 208.0) Aperture PA Range 233.92542306 to 235.92542306 Degrees (V3 234.0 to 236.0) Aperture PA Range 240.92542306 to 262.92542306 Degrees (V3 241.0 to 263.0) Aperture PA Range 275.92542306 to 285.92542306 Degrees (V3 276.0 to 286.0) Aperture PA Range 331.92542306 to 349.92542306 Degrees (V3 332.0 to 350.0) Visits Same PA

Observation

Proposal 5114, Observation 11: N159S NIRCam

Wed Sep 11 22:00:20 GMT 2024

Diagnostic Status: Warning

Observing Template: NIRCam Imaging

Diagnostics

(Visit 11:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(9)	LHA-120-N-159S-NIRCam	RA: 05 40 4.7000 (85.0195833d) Dec: -69 50 30.60 (-69.84183d) Equinox: J2000	Epoch of Position: 2000	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.
Category=Stellar Cluster
Description=[Protoclusters, Young star clusters]

Template

Module	Subarray	Target Placement
ALL	FULL	Module Gap

Dithers

#	Primary Dither Type	Primary Dithers	Subpixel Dither Type	Dither Size	Subpixel Positions
1	NONE		SMALL-GRID-DITHER		4

Spectral Elements

#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	ETC Wkbk.Calc ID
1	F212N	F470N+F444W	BRIGHT2	6	1	4	4	515.365	
2	F187N	F460M	BRIGHT2	5	1	4	4	429.471	
3	F210M	F335M	BRIGHT2	5	1	4	4	429.471	
4	F140M	F360M	BRIGHT2	5	1	4	4	429.471	
5	F115W	F300M	BRIGHT2	5	1	4	4	429.471	

Special Requirements

Offset -96.18637491236724 arcsec, -2.9445691656558384 arcsec

Observation

Proposal 5114, Observation 12: N159-MIRI

Wed Sep 11 22:00:20 GMT 2024

Diagnostic Status: Warning

Observing Template: MIRI Imaging

Background Observations:[N159-OFF MIRI (Obs 7)]

Diagnostics

(N159-MIRI (Obs 12)) Warning (Form): Target requiring background exposure selected for template that doesn't require background exposure

(Visit 12:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

(Visit 12:2) Warning (Form): Overheads are provisional until the Visit Planner has been run.

(Visit 12:3) Warning (Form): Overheads are provisional until the Visit Planner has been run.

(Visit 12:4) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(8)	LHA-120-N-159-MIRI	RA: 05 39 44.9700 (84.9373750d) Dec: -69 45 58.60 (-69.76628d) Equinox: J2000	Epoch of Position: 2000	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.
Category=Stellar Cluster
Description=[OB associations, Protoclusters, Young star clusters]

Template

Subarray
FULL

Mosaic

Rows	Columns	Row Overlap %	Column Overlap %	Row shift (deg)	Column shift (deg)	Tile Order
2	2	10.0	10.0	-15.0	-45.0	DEFAULT

Dithers

#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size
1	CYCLING	1	4		5	1			SMALL

Spectral Elements

#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	F770W	FASTR1	28	1	1	Dither 1	4	4	310.804	
2	F1000W	FASTR1	36	1	1	Dither 1	4	4	399.606	
3	F1280W	FASTR1	28	1	1	Dither 1	4	4	310.804	
4	F1500W	FASTR1	32	1	1	Dither 1	4	4	355.205	
5	F2100W	FASTR1	51	1	1	Dither 1	4	4	566.108	

Special Requirements

Group Visits within 53.0 Days
Aperture PA Range 257.83544897 to 258.83544897 Degrees (V3 253.0 to 254.0)
Visits Same PA

Group Observations 7, 9, 12, Non-interruptible

Observation

Proposal 5114, Observation 22: N159-MIRI

Wed Sep 11 22:00:20 GMT 2024

Diagnostic Status: Warning

Observing Template: MIRI Imaging

Background Observations:[N159-OFF MIRI (Obs 17)]

Comments: Repeat of failed visit 12:2

Diagnostics

(N159-MIRI (Obs 22)) Warning (Form): Target requiring background exposure selected for template that doesn't require background exposure

(Visit 22:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(10)	LHA-120-N-159-MIRI-12.2-repeat	RA: 05 39 48.7000 (84.9529167d) Dec: -69 46 15.80 (-69.77106d) Equinox: J2000	Epoch of Position: 2000	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.
Category=Stellar Cluster
Description=[OB associations, Protoclusters, Young star clusters]

Template

Subarray
FULL

Dithers

#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size
1	CYCLING	1	4		5	1			SMALL

Spectral Elements

#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	F770W	FASTR1	28	1	1	Dither 1	4	4	310.804	
2	F1000W	FASTR1	36	1	1	Dither 1	4	4	399.606	
3	F1280W	FASTR1	28	1	1	Dither 1	4	4	310.804	
4	F1500W	FASTR1	32	1	1	Dither 1	4	4	355.205	
5	F2100W	FASTR1	51	1	1	Dither 1	4	4	566.108	

Special Requirements

Aperture PA Range 66.83544897 to 78.83544897 Degrees (V3 62.0 to 74.0)
Aperture PA Range 243.83544897 to 260.83544897 Degrees (V3 239.0 to 256.0)

Group Observations 17, 22, Non-interruptible

Observation

Proposal 5114, Observation 9: N159S-MIRI Wed Sep 11 22:00:20 GMT 2024

Diagnostic Status: Warning

Observing Template: MIRI Imaging

Background Observations:[N159-OFF MIRI (Obs 7)]

Diagnostics

(N159S-MIRI (Obs 9)) Warning (Form): Target requiring background exposure selected for template that doesn't require background exposure

(Visit 9:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

(Visit 9:2) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(6)	LHA-120-N-159S-MIRI	RA: 05 40 4.7000 (85.0195833d) Dec: -69 50 30.60 (-69.84183d) Equinox: J2000	Epoch of Position: 2000	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.
Category=Stellar Cluster
Description=[Protoclusters, Young star clusters]

Template

Subarray
FULL

Mosaic

Rows	Columns	Row Overlap %	Column Overlap %	Row shift (deg)	Column shift (deg)	Tile Order
1	2	10.0	10.0	0.0	0.0	DEFAULT

Dithers

#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size
1	CYCLING	1	4		5	1			SMALL

Spectral Elements

#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	F770W	FASTR1	28	1	1	Dither 1	4	4	310.804	
2	F1000W	FASTR1	36	1	1	Dither 1	4	4	399.606	
3	F1280W	FASTR1	28	1	1	Dither 1	4	4	310.804	
4	F1500W	FASTR1	32	1	1	Dither 1	4	4	355.205	
5	F2100W	FASTR1	51	1	1	Dither 1	4	4	566.108	

Special Requirements

Group Visits within 53.0 Days
Visits Same PA

Group Observations 7, 9, 12, Non-interruptible

Observation

Proposal 5114, Observation 7: N159-OFF MIRI Wed Sep 11 22:00:20 GMT 2024

Diagnostic Status: Warning

Observing Template: MIRI Imaging

Background Observation For: [N159S-MIRI (Obs 9), N159-MIRI (Obs 12)]

Diagnostics

(Visit 7:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(4)	OFF-FIELD-MIRI	RA: 05 41 52.0000 (85.4666667d) Dec: -69 38 12.00 (-69.63667d) Equinox: J2000		

Comments:
Category=Stellar Cluster
Description=[OB associations, Protoclusters, T associations, Young star clusters]

Template

Subarray
FULL

Dithers

#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size
1	CYCLING	1	4		5	1			SMALL

Spectral Elements

#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	F770W	FASTR1	28	1	1	Dither 1	4	4	310.804	
2	F1000W	FASTR1	36	1	1	Dither 1	4	4	399.606	
3	F1280W	FASTR1	28	1	1	Dither 1	4	4	310.804	
4	F1500W	FASTR1	32	1	1	Dither 1	4	4	355.205	
5	F2100W	FASTR1	51	1	1	Dither 1	4	4	566.108	

Special Requirements

Group Observations 7, 9, 12, Non-interruptible

Observation

Proposal 5114, Observation 17: N159-OFF MIRI Wed Sep 11 22:00:20 GMT 2024

Diagnostic Status: Warning

Observing Template: MIRI Imaging

Background Observation For: [N159-MIRI (Obs 22)]

Comments: Repeat of failed visit 7:1.

Diagnostics

(Visit 17:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(4)	OFF-FIELD-MIRI	RA: 05 41 52.0000 (85.4666667d) Dec: -69 38 12.00 (-69.63667d) Equinox: J2000		

Comments:
Category=Stellar Cluster
Description=[OB associations, Protoclusters, T associations, Young star clusters]

Template

Subarray
FULL

Dithers

#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size
1	CYCLING	1	4		5	1			SMALL

Spectral Elements

#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	F770W	FASTR1	28	1	1	Dither 1	4	4	310.804	
2	F1000W	FASTR1	36	1	1	Dither 1	4	4	399.606	
3	F1280W	FASTR1	28	1	1	Dither 1	4	4	310.804	
4	F1500W	FASTR1	32	1	1	Dither 1	4	4	355.205	
5	F2100W	FASTR1	51	1	1	Dither 1	4	4	566.108	

Special Requirements

Group Observations 17, 22, Non-interruptible